

Machine Learning - CSTE 5131

Machine Learning: An Overview

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Welcome to CSTE 5131

Year - 1 Term - 1 (2024)

- Textbook & Lecture Notes:

- **Textbook:**

- “*Data Mining: Concepts and Techniques*”, J. Han, M. Kamber, and J. Pei, Morgan Kaufmann (3rd Edition)
 - “*Machine Learning*” by Tom Mitchell, McGraw-Hill

- **Lecture Notes by Andrew Ng:**

- http://cs229.stanford.edu/lectures-spring2022/main_notes.pdf

- **Strongly Recommended:** Class lectures (available after each class)

Evaluation



- Exam - 70%
- Assignment/Class Test/Presentation: 25%
- Attendance: 5%



Web Resource Examples

- General Data Mining (DM)/ML Site: <http://www.kdnuggets.com>
 - A collection of DM/ML publications, software/tools, data repositories, companies, etc.
- Data Mining and Machine Learning Software/Tools/Programming Language –
Weka: <http://www.cs.waikato.ac.nz/ml/weka/>
 - A collection of machine learning algorithms for solving real-world data mining problems. It is written in Java and runs on almost any platform.
R: <https://cloud.r-project.org/>
Python: <https://www.python.org/>
- Database Repository: <http://www.ics.uci.edu/~mlearn/MLRepository.html>
<https://www.kaggle.com/>
<https://datasetsearch.research.google.com/>
<https://visualdata.io/discovery>

Overview of Machine Learning (ML)



- Machine Learning Definition:

Arthur Samuel (1959). Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.



Overview of Machine Learning (ML)?

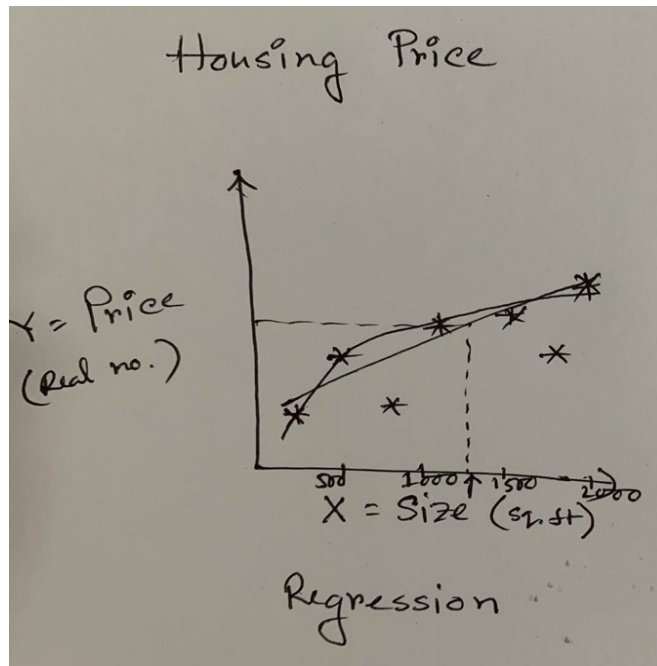
- Learn from data: identifying patterns in the data, build a model helping to make decisions with minimal human intervention
- Supervised learning
 - build a model based on inputs that contain both the data and some model values
 - classification and regression
- Unsupervised learning
 - build a model from the data alone, like grouping or clustering of data points
 - clustering



Overview of Machine Learning (ML)



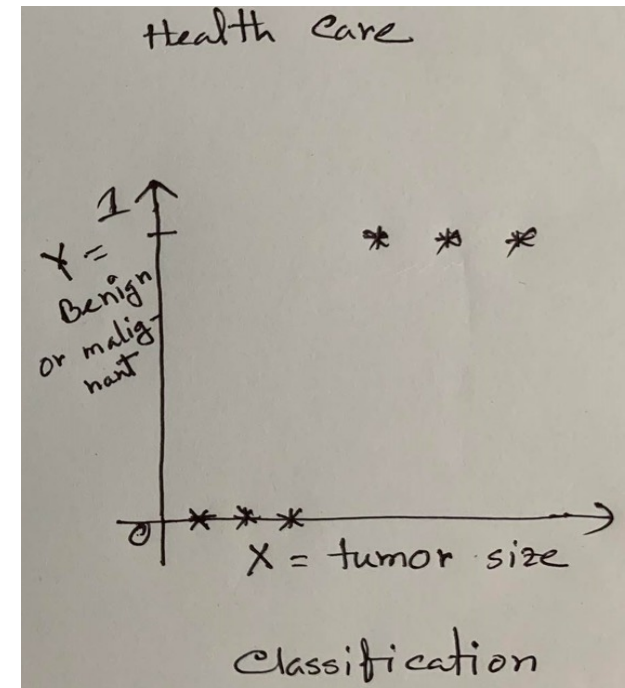
- Supervised learning



Dataset (X, Y)

Input: X, Output: Y

Goal: $X \rightarrow Y$ (Learn a mapping from X to Y)



Supervised learning

Data

Input: X

Features/Variables

Input: X, Output/Label: Y

Goal: $X \rightarrow Y$ (Learn a mapping from X to Y)

Output/Class
label: Y

Attributes				Classes
Gender	Car ownership	Travel Cost (\$)/km	Income Level	Transportation mode
Male	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Female	1	Cheap	Medium	Train
Female	0	Cheap	Low	Bus
Male	1	Cheap	Medium	Bus
Male	0	Standard	Medium	Train
Female	1	Standard	Medium	Train
Female	1	Expensive	High	Car
Male	2	Expensive	Medium	Car
Female	2	Expensive	High	Car

Supervised learning

Classifying Upon New Data Input

Person name	Gender	Car ownership	Travel Cost (\$)/km	Income Level	
Alex	Male	1	Standard	High	→
Buddy	Male	0	Cheap	Medium	→
Cherry	Female	1	Cheap	High	→

Input Data

Bus/Train/Car?

Output/Class label

- Based upon the model created from the training data, the machine is able to classify into predefined classes, which in this case are bus or train or car.

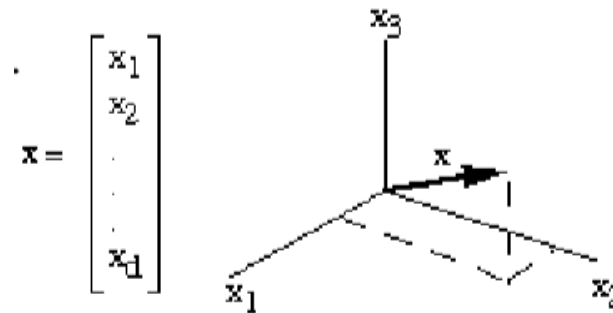


Attributes/Features/Properties

- In machine learning or pattern Recognition a **feature** is an individual measurable heuristic property of a phenomenon being observed
 - **Ex:** lightness, length, width, color of an object
- Choosing **discriminating and independent** features is key to any pattern recognition problem
- A set of features of a given data instance is often grouped into a feature vector

Feature Vector

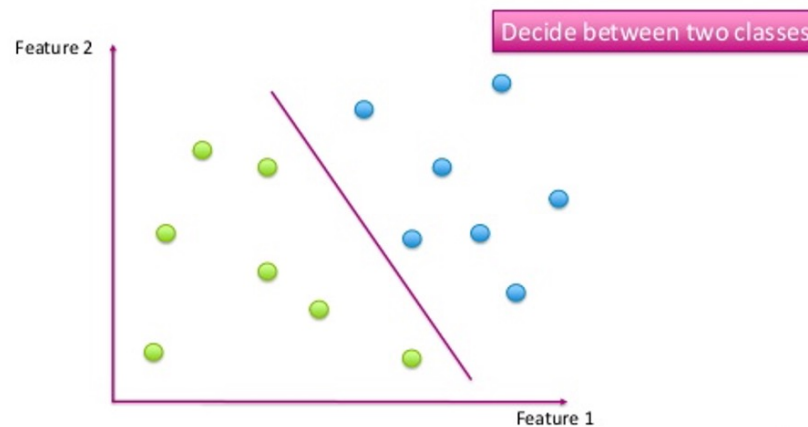
- Usually, a single object can be represented using several features, e.g.
 - x_1 = shape
 - x_2 = size
 - x_3 = color
- **X** becomes a feature vector
 - x is a point in a d-dimensional feature space.



Feature Space

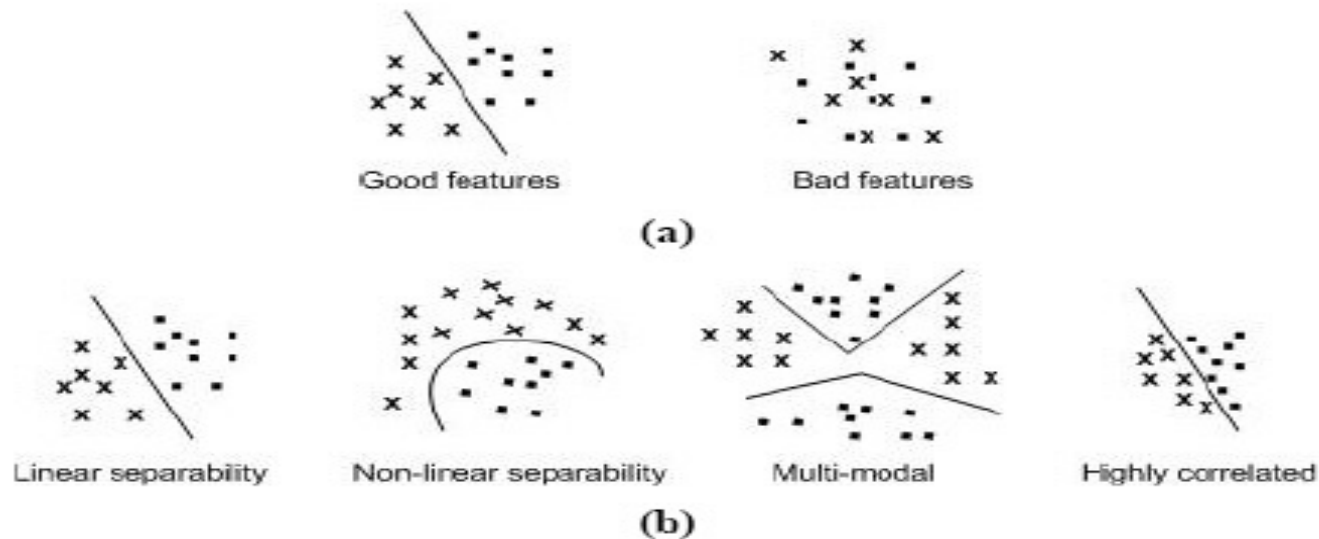
- The **D-dimensional space** defined by a feature vector is known as a feature space.
 - Objects are represented as points in feature space

Classification model



Decision Boundary

- The goal of a classifier is to **partition the feature space** into class-labeled decision regions.
- **Borders** between decision regions are called decision boundaries



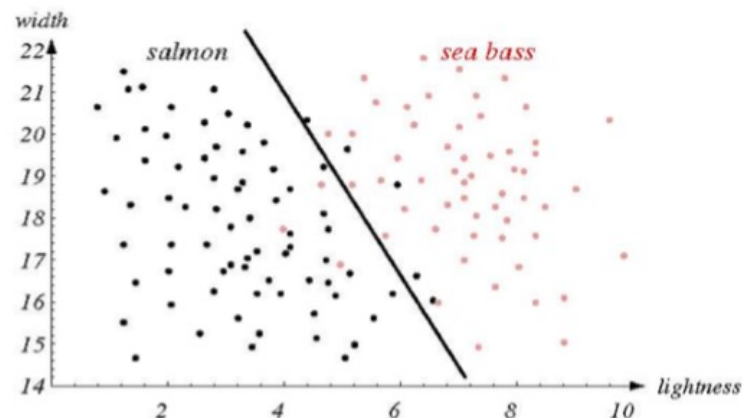


A Case Study: Fish Classification

- Multiple Features :
 - Single features might not yield the best performance.
 - To improve recognition, we might have to use more than one feature at a time.
 - Combinations of features might yield better performance.
 - Assume we also observed that sea bass are typically wider than salmon.

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \begin{matrix} x_1 : \text{lightness} \\ x_2 : \text{width} \end{matrix}$$

Each fish image is now represented by a point in this 2D feature space



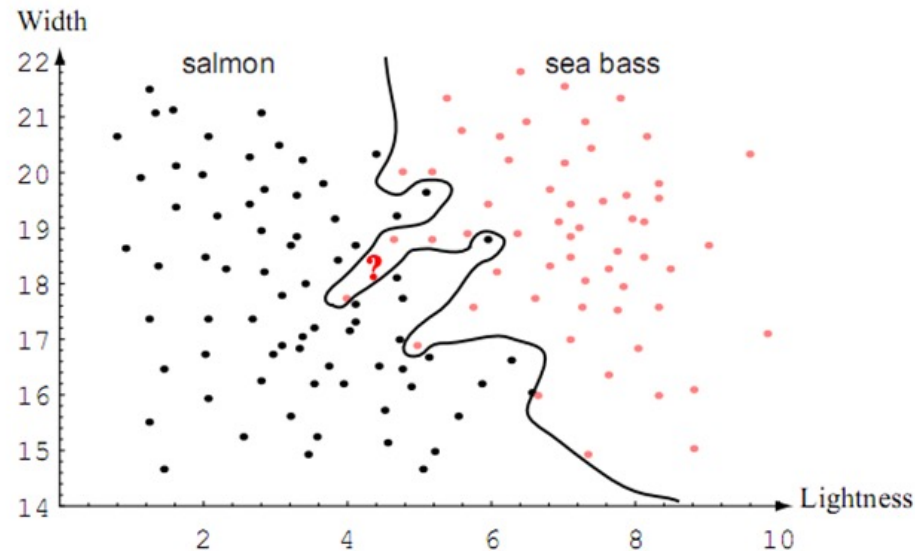
Scatter plot of lightness and width features for training samples. We can draw a decision boundary to divide the feature space into two regions. Does it look better than using only lightness?



Designing a Classifier - Priority on Generalization

- If a model is **extremely complicated**, the classifier would have a decision boundary more complex than the simple straight line, but the training pattern would be separated/classified perfectly

We may distinguish training samples perfectly but how can we predict how well we can generalize to unknown samples?



DANGER OF OVER FITTING!!

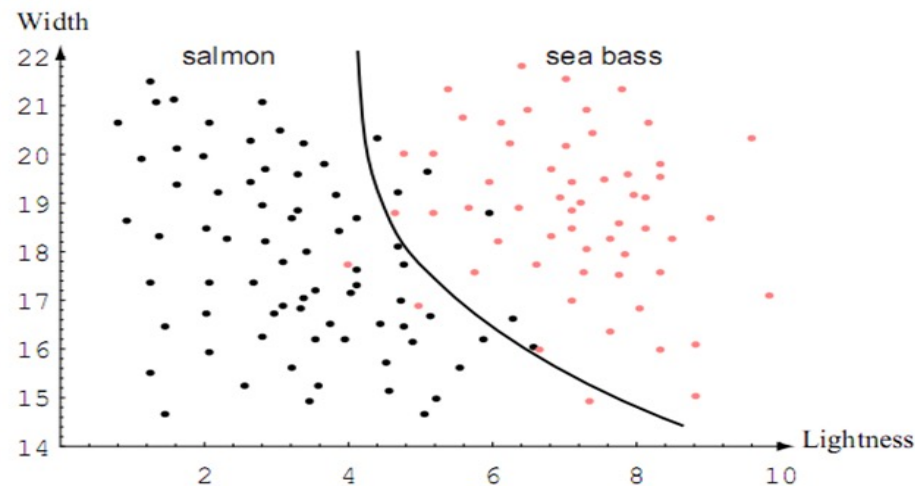
CLASSIFIER WILL FAIL TO
GENERALIZE TO NEW
DATA...

****Bias-variance trade-off**



Designing a Classifier - Priority on Generalization

- How can we manage the tradeoff between complexity of decision rules and their performance to unknown samples
- Thus, the central aim of designing a classifier should be to **seek for simplicity yet that would perform well enough** for Nobel patterns that is one capable of acting accurately based on a wide variety of sample instances



This is the prime idea behind **generalization**

Courtesy: Dr. Fahim Irfan Alam



Machine Learning Models

- K-Nearest Neighbor
- Decision Tree
- Random Forest
- Support Vector Machine
- Naïve Bayes
- Linear Regression
- Boosting Algorithms – AdaBoost, Gradient Boost, XGBoost
- Neural Network and Deep Learning – MLP, CNN, LSTM
- AND MANY MORE.....

Big Data





Big Data

- How big is it?
- How to get useful information from data?
 - 1.2 trillion searches, 146 languages. What did the world search for in 2012?
 - 175 millions registered voters, a million none registered voters, “How Obama Knew How You'd Vote, Even Before You Did?”
 - 350 million photos and videos sent by snap chatters every day (2013).



Figure 1.2 The world is data rich but information poor.

How to get useful info from data?



<https://www.youtube.com/watch?v=-vJ3ycpvk4I>

Big Data Applications



Thank you

